DELIVERY-AWARE SPARSE CAUSAL ROUTING FOR DISTRIBUTED MULTI-OBJECT TRACKING

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ABSTRACT

We study sparse communication for distributed labeled multi-Bernoulli tracking when mobile formations change their physical neighbors. A causal routing policy retains a feasible formation tree, repairs it only when necessary, and combines local directed cycles with bidirectional gateways. The scheduled backbone uses $N + 2(F - 1)$ posterior messages for N sensors in F formations. We separate planned connectivity from delivered, label-specific fusion. Paired 160-step development experiments with 24 and 36 sensors reduce attempted posterior bytes by 10.0% and 21.8% relative to fixed routing. Mean OSPA decreases by 2.40% and 0.37%, respectively; larger conditional localization gains do not imply comparable target-set recovery. A post-hoc error budget shows that fixed-cardinality localization improvements alone can reduce mean OSPA by at most 0.28 and 1.31 m in the sparse arms. A receiver ablation exposes a precision–consistency tradeoff.

Index Terms— Multi-object tracking, labeled multi-Bernoulli, distributed fusion, dynamic topology, packet loss

1. INTRODUCTION

Mobile formations exchange posteriors to track beyond individual fields of view. Motion breaks routes; redundant inputs consume bandwidth. Which inputs should reach which receivers soon enough to support both object existence and location?

The labeled multi-Bernoulli (LMB) filter represents object existence and spatial states [1]. Consensus LMB filtering uses Kullback–Leibler averaging (KLA), or normalized geometric fusion [2, 3]. Unlike repeated consensus on fixed inputs, tracking interleaves observations with few communication rounds; scheduled connectivity guarantees neither delivery nor lower error.

Randomized gossip relates averaging time to a stochastic matrix spectrum [4]; event-triggered Bayesian filtering uses posterior divergence from a predicted last transmission [5]. Ramachandran et al. jointly reconfigure topology and fusion weights after sensor-quality degradation, then reposition robots for coverage [6]. We do not claim route repair as new. With prescribed motion, we study sparse messages, directed losses and finite-round LMB fusion under unequal views.

Multi-view GCI includes GM-PHD clustering and compensation [7], LMB fusion under unequal views [8], and centralized per-label information-based weighting [9]. Our arms share a repository-specific absent-label FoV censor, not these complete methods; routing uses physical links, not learned weights.

Label-matched GCI [10] and LMB minimum information loss fusion [11] address label correspondence and the pooling objective, respectively. Unequal-view fusion also treats exclusive tracks [12]. These are separate estimator baselines, not components of the routing arms compared here.

We give exact message counts and packet-induced weight bounds, and compare full with sparse repair. One opened seed per scale reveals development tradeoffs, not generalization.

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2. SPARSE CAUSAL LMB ROUTING

2.1. Fusion and communication model

Sensor i maintains an LMB posterior $\pi_i = \{(r_i^\ell, p_i^\ell)\}_{\ell\in\mathbb{L}_i}$, where r_i^ℓ is the existence probability and $p_i^\ell(x)$ is a normalized spatial density. For a common active label and weights $w_j \geq 0$, $\sum_j w_j = 1$, exact KLA satisfies [3]

$$\eta^\ell = \int \prod_j p_j^\ell(x)^{w_j}\, dx, \qquad \bar{p}^\ell(x) = \frac{\prod_j p_j^\ell(x)^{w_j}}{\eta^\ell}, \tag{1}$$

$$\operatorname{logit} \bar{r}^\ell = \sum_j w_j \operatorname{logit} r_j^\ell + \log \eta^\ell. \tag{2}$$

For these exact common-label densities, Hölder's inequality gives $\eta^\ell \leq 1$: spatial disagreement can lower existence log odds relative to their weighted input value. Thus a new route changes both information access and

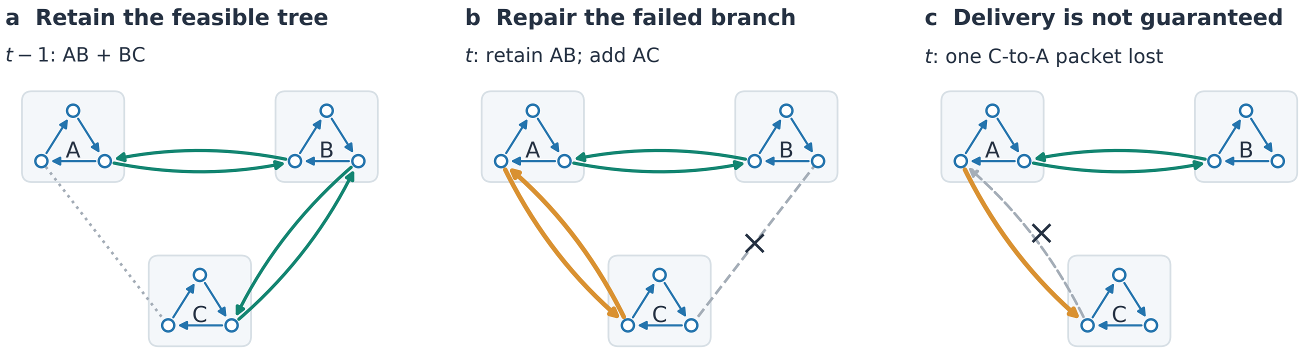

Fig. 1. Sparse causal repair with three illustrative formations of three sensors (not an M24/X36 layout). Blue arrows are local cycles. (a) Retain green AB–BC; dotted AC is unused. (b) Replace failed BC by orange AC, retaining AB. Both plans use 13 messages and at most two nonself inputs per receiver. (c) One C-to-A loss leaves 12 delivered messages and no path out of C. Scheduled connectivity therefore does not ensure delivery, let alone label-specific fusion quality.

density compatibility; more mixing need not preserve the target set.

Let G_t^{phys} describe current physical links, and let A_t be the scheduled directed graph, with rows denoting receivers and columns senders. The controller observes current positions, link reliabilities and its previous route, but no target truth or future trajectory. The implemented controller uses network-wide current geometry; decentralized acquisition and dissemination of this control information are outside the measured posterior-payload budget. Fusion itself is performed at each sensor.

2.2. Causal tree repair and sparse inputs

Sensors are partitioned into F formations. A candidate edge in the formation graph is usable when current physical sensor pairs can realize the required gateways. The policy first tries the preceding formation tree. When that tree or its gateway assignment is infeasible, it selects a current feasible tree lexicographically: minimize replaced formation edges, maximize bottleneck reliability, maximize total log reliability, then minimize distance. Gateway feasibility includes the receiver input constraints. Local cycles and gateway endpoints are rebuilt from current geometry; only the formation tree is retained (Fig. 1).

The sparse policy retains one directed cycle inside each formation. Each undirected edge of the formation tree is realized by one directed gateway in each direction; the two directions need not use the same sensor pair. Every sensor receives one dominant local input, and gateway receivers also receive one cross-formation input. The planned weights are 0.70 for the dominant input and 0.05 for a retained residual input, with the remaining mass assigned to self. The full-repair reference instead retains two nonself inputs per receiver. Removing its unused local residual input increases the sparse receiver's self weight by 0.05.

Proposition 1 (scheduled backbone). If every local cycle and every directed gateway is physically feasible, the scheduled sensor graph is strongly connected and contains exactly

$$M = N + 2(F - 1) \tag{3}$$

directed messages per round. Proof. The local cycles contribute N edges and provide directed paths between sensors in each formation. The tree contributes $F-1$ formation edges, each realized in both directions. Concatenating local paths and gateways gives a directed path between any two sensors. The edge sets are disjoint, giving (3). This is a count for the chosen architecture, not a global lower bound over all strongly connected directed graphs or a guarantee about delivered links.

3. PACKET DELIVERY AND FINITE-ROUND FUSION

3.1. Where a missing weight goes

Let W be a planned row-stochastic matrix with positive diagonal. For receiver i, let $D_{ij} \in \{0,1\}$ indicate delivery, set $D_{ii}=1$, and define missing weight mass $m_i = \sum_{j:D_{ij}=0} W_{ij}$. The current reference implementation omits an unavailable input and uses

$$R_{ij} = \frac{W_{ij} D_{ij}}{1 - m_i}. \tag{4}$$

This differs from the planned removal of a local edge. An existing alternative receiver rule returns missing mass to self:

$$S_{ij} = W_{ij} D_{ij} + m_i \mathbf{1}\{i = j\}. \tag{5}$$

Both are row stochastic, and both satisfy $\|R_{i:} - W_{i:}\|_1 = \|S_{i:} - W_{i:}\|_1 = 2m_i$. Their distinction is therefore not a smaller row-wise perturbation norm. Rule (5) preserves the planned weight of each surviving neighbor. If a 0.70-weight dominant input is missing, a surviving 0.05-weight input becomes $0.05/0.30 = 0.1667$ under (4), but stays at 0.05 under (5). The latter avoids amplification but can slow useful information transfer.

Neither rule preserves double stochasticity under arbitrary directed losses. For example, a symmetric two-node matrix with off-diagonal weight a becomes $\left[\begin{smallmatrix} 1 & 0 \\ a & 1-a \end{smallmatrix}\right]$ after one direction fails under self fallback. Its column sums are $1+a$ and $1-a$. Reciprocal scheduled gateways alone consequently do not establish equal-weight KLA consensus after packet loss.

3.2. A finite-round sensitivity bound

Freeze positive densities q_j on a common support between fusion rounds, and write $q_\alpha \propto \prod_j q_j^{\alpha_j}$ for a probability vector α. Assume $\operatorname{osc}(\log q_j) = \sup \log q_j - \inf \log q_j \leq L$. Let $P = \widetilde{W}_K \cdots \widetilde{W}_1$ and $Q = W_K \cdots W_1$ be delivered and planned products. For either receiver rule,

$$
\begin{aligned}
D_{\mathrm{KL}}(q_{P_{i:}} \| q_{Q_{i:}}) &\leq L \|P_{i:} - Q_{i:}\|_1 \\
&\leq 2L \sum_{k=1}^{K} \max_j m_{j,k}. \qquad (6)
\end{aligned}
$$

To see this, $h = \sum_j (\alpha_j - \beta_j) \log q_j$ has oscillation at most $L\|\alpha - \beta\|_1$, and the log normalization ratio lies between $\inf h$ and $\sup h$. Telescoping the products, whose stochastic factors have infinity norm one, bounds their row difference by $\sum_k \|\widetilde{W}_k - W_k\|_\infty = 2\sum_k \max_j m_{j,k}$. This elementary bound favors neither receiver rule. Unbounded Gaussian support, mixture truncation, changing labels and intervening observations prevent its direct use as a guarantee for the recursive tracker.

4. EXPERIMENTS AND REMAINING TRADEOFFS

4.1. Paired protocol

M24 and X36 use four and six formations of six sensors, tracking 16 and 24 targets over 160 steps. In this formation-braid scene, target handovers overlap physical-route failures. Formation members share a heading, 120° fields of view and 300 m range. Nominal detection probability, clutter rate and noise standard deviation are 0.88, 4 and 7 m; sensing degrades with range and off-axis angle. Each scale uses seed 1301, with truth, measurements, directed delivery uniforms and filter randomness paired.

The fixed baseline keeps its initial formation tree, while recomputing sensor gateways when that tree remains feasible. Otherwise, it retains the still-physical edges of its initial sensor route. Full causal repair retains two inputs per receiver and changes the formation tree when required. Sparse causal repair uses (3). These three arms share packet renormalization (4). A fourth X36 arm changes only missing-input handling to (5), including unavailable or empty neighbor packets.

All arms share a guarded powered-GM approximation: at most three components per input, eight fused components and 256 tuples; overlap above one invokes moment-matched Gaussian fallback. Arbitrary mixture powers are not exact. Component-mean detection probabilities ignore FoV boundary-crossing mass; boundary-aware splitting changes the local spatial update [13]. Present labels are not excluded solely for being outside the view. Each step uses one fusion round and shared birth labels without matching. MAP cardinality determines the output; other labels remain unless pruned below existence 0.01.

Metrics are Euclidean position OSPA (E-OSPA, order 2, cutoff 150 m) [14] and Hungarian-matched position RMSE, averaged over sensor–time cells. RMSE excludes nonempty–empty comparisons and must be read with absolute cardinality error. Focus consistency is pairwise E-OSPA among one fixed representative per formation over $t = 40$–140, not all-node agreement. Attempted posterior bytes include failed transmissions but exclude routing-control traffic.

4.2. What the comparisons establish

Table 1 and Fig. 2 separate repair from sparsification. Full causal repair improves the three reported error measures over fixed routing at both scales, but increases attempted bytes by 10.052% on M24 and 5.708% on X36. The sparse policy instead saves 10.041% and 21.830% of attempted bytes. On M24 its E-OSPA and conditional RMSE improve by 2.403% and 46.190%, and focus consistency improves by 1.449%. However, the worst formation-wise RMSE change is a 24.085% degradation.

On X36 sparse repair improves E-OSPA by only 0.368% despite a 47.655% conditional-RMSE reduction. Mean absolute cardinality error rises from 18.456 to 18.597, and focus consistency worsens by 0.776%. The RMSE decrease therefore does not establish a comparable improvement in complete target-set recovery.

X36 sparse repair has finite RMSE on 97.465% of sensor–time cells, versus 100% for fixed routing. On common finite cells the means are 19.329 and 31.394 m, a 38.431% reduction. Matched targets can still differ.

Table 1. Paired 160-step development episodes, seed 1301. All error and payload measures are lower-is-better. Errors are in metres; payload is attempted posterior traffic in decimal MB. Message counts precede delivery.

Scale	Routing policy	E-OSPA	Cond. RMSE	Focus consistency	Payload	Messages/step
M24	Fixed formation tree	125.478	22.640	133.599	40.769	46–48
	Full causal repair	122.380	14.081	131.913	44.867	48
	Sparse causal repair	122.462	12.183	131.664	36.676	30
X36	Fixed formation tree	132.680	36.925	139.407	76.871	66–72
	Full causal repair	131.795	19.586	139.004	81.259	72
	Sparse causal repair	132.192	19.329	140.489	60.090	46
	Sparse + self fallback	131.961	20.263	139.273	60.317	46

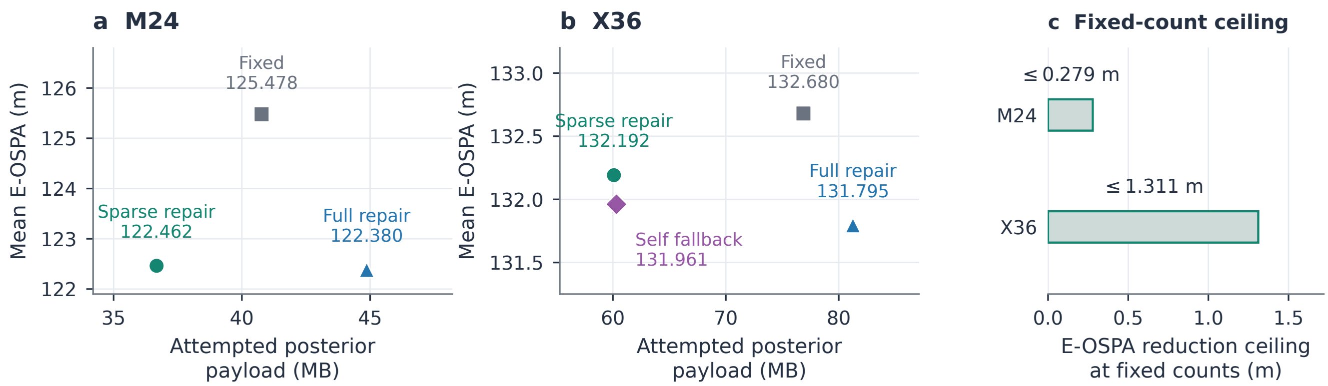

Fig. 2. Communication savings and the remaining target-set gap. (a,b) Payload versus E-OSPA for Table 1; lower left is better, with different vertical scales. Each point is one development episode, without across-seed uncertainty. (c) Upper bounds on further E-OSPA reduction from localization alone in the sparse arms, holding output counts fixed. These bounds are not achieved gains or confidence intervals.

4.3. How much can localization alone improve?

To separate position refinement from target-set recovery, write squared OSPA at one sensor–time cell as

$$E^2 = C + L, \qquad C = c^2 \frac{|n - T|}{\max(n, T)}, \qquad L \geq 0, \qquad (7)$$

where T, n are true and estimated counts and $c = 150\,\mathrm{m}$; two empty sets give $E = C = L = 0$. Perfect localization at fixed counts leaves $\sqrt{C}$. Saved absolute count errors admit two signs. OSPA's floor and RMSE matching exclude infeasible counts, leaving 0/3840 ambiguous sparse M24 cells and 121/5760 X36 cells with $C_{\min} \leq C \leq C_{\max}$. For sparse repair, $\sum C / \sum E^2$ is 99.567% on M24 and 98.074–99.337% on X36, not a linear decomposition of mean OSPA. The remaining fixed-count gain is at most mean$(E - \sqrt{C_{\min}})$: 0.279 and 1.311 m. These ceilings, shown in Fig. 2(c), are not achieved gains or proof of correct target identities. Even perfect fixed-count localization leaves the count penalty; substantial gains require recovering the target set.

The delivered sparse graph is strongly connected in only 41/160 M24 and 18/160 X36 steps; these counts measure neither label-wise nor temporal connectivity.

4.4. Receiver ablation and limitations

Against sparse renormalization, self fallback improves X36 E-OSPA by 0.175% and focus consistency by 0.865%; cardinality error drops to 18.518. Conditional RMSE worsens by 4.836%, with a 37.361% loss in the most affected formation. Neither dominates.

Future comparisons must separate estimator changes from routing effects. Multi-view and label-matched baselines, independent seeds and layouts, control traffic and decentralized routing remain open.

5. CONCLUSION

Sparse repair saves payload but retains target-set and formation-wise tradeoffs. Planned connectivity does not ensure accurate target recovery.

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